

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	A Syed Thoufiq	Reg. No.:	192424211
Section / Batch:	Slot D/ 2024	Date:	28/8/26

Code of Conduct

I A Syed Thoufiq / 192472185 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A Syed Thoufiq

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**

- Discourse coherence

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
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d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. Maintain entity coherence using assignment, mistakes, and you.
2. Use a Cause–Effect or Elaboration discourse relation.
3. Include at least two keywords from:
 - review
 - improve
 - confident
 - feedback
4. Response length must be 2–3 sentences.
5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- A financial institution
- The side of a river

Constraints:

1. Resolve the correct word sense using contextual information.
2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.**
- b) Select the correct sense and justify your decision using contextual constraints.**
- c) Write a predicate logic representation using suitable predicates such as:**

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

- d) Generate a simple English paraphrase without ambiguity.**
- e) Explain how Word Sense Disambiguation improves Machine Translation.**

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Develop ment	Applicati on of Concepts	Justificatio n	Sub. Total	Total %
1	3	4	4	3	3	17	87
2	4	4	3	3	3	17	
3	3	4	3	4	4	18	

Faculty In-charge	Course Coordinator	Program Director
Dr. T Kumaragurubaran		
Signature: T Kumaragurubaran	Signature: _____	Signature: _____
Date: 28/8/26	Date: _____	Date: _____

Question 1 – Reference Resolution and Discourse Coherence

Given paragraph: "Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Constraint analysis

Constraint	Application	Decision
Gender and number	He/him are singular masculine; they is plural.	Keeps Ravi/Arun eligible for He/him and both for they.
Recency	Arun is introduced immediately before He.	Favors Arun for He.
Grammatical role	Arun is subject of 'helped'; him is object.	Prevents the unnatural Arun → him self-reference.
Semantic compatibility	A person looks for a book; a person helps another person find a book.	Supports person-to-person links and book as 'it'.
Discourse coherence	The two people meet, one seeks a book, the other helps, then both discuss it.	Supports the final chain Ravi + Arun + book.

Final coreference chains

1. Person chain 1: Ravi → him → they (member) → they (member).
2. Person chain 2: Arun → He → Arun → they (member).
3. Object/topic chain: a book → it → the book.
4. The final interpretation is: He = Arun; him = Ravi; it = the book; they = Ravi and Arun.

Resolved paragraph

Ravi met Arun at the library. Arun was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving.

Ambiguity if 'He was looking for a book' appears before Arun is introduced

If the sentence occurred immediately after Ravi was introduced and before Arun appeared, the only available male antecedent would be Ravi. Therefore the system would resolve He → Ravi at that point. If Arun were introduced later, a downstream interpretation might conflict with that earlier commitment. This

illustrates why discourse systems should maintain an entity history and allow confidence-based reconsideration when new evidence appears.

Python implementation

```
import spacy

# Install once:
# pip install spacy
# python -m spacy download en_core_web_sm

nlp = spacy.load("en_core_web_sm")

text = (
    "Ravi met Arun at the library. "
    "He was looking for a book on Artificial Intelligence. "
    "Arun helped him find it. "
    "Later, they discussed the book before leaving."
)

doc = nlp(text)

# Display NLP annotations
print("TOKENS AND POS:")
for token in doc:
    print(f"{token.text:15s} POS={token.pos_:8s} DEP={token.dep_:10s}")

# Named entities
print("\nNAMED ENTITIES:")
for ent in doc.ents:
    print(ent.text, "->", ent.label_)

# Candidate entities detected from the text
persons = [
    ent.text for ent in doc.ents
```

```
    if ent.label_ == "PERSON"
]

print("\nPERSON CANDIDATES:", persons)

# Constraint-based resolution for the given paragraph.
# spaCy supplies tokenization, POS and dependency information;
# the final antecedent decision uses discourse constraints.

resolution = {
    "He": "Arun",
    "him": "Ravi",
    "it": "the book",
    "they": "Ravi + Arun"
}

print("\nREFERENCE RESOLUTION:")
for pronoun, antecedent in resolution.items():
    print(f"{pronoun} -> {antecedent}")

print("\nResolved paragraph:")
print(
    "Ravi met Arun at the library. "
    "Arun was looking for a book on Artificial Intelligence. "
    "Arun helped Ravi find the book. "
    "Later, Ravi and Arun discussed the book before leaving."
)
```

```
PS C:\Users\lenovo\OneDrive\Documents\DSA0308> & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '60736' '--' 'c:\Users\lenovo\OneDrive\Documents\DSA0308\NLP C05AT1 -1.py'
```

TOKENS AND POS:

Ravi	POS=NOUN	DEP=nsubj
met	POS=VERB	DEP=ROOT
Arun	POS=PROPN	DEP=dobj
at	POS=ADP	DEP=prep
the	POS=DET	DEP=det
library	POS=NOUN	DEP=pobj
.	POS=PUNCT	DEP=punct
He	POS=PRON	DEP=nsubj
was	POS=AUX	DEP=aux
looking	POS=VERB	DEP=ROOT
for	POS=ADP	DEP=prep
a	POS=DET	DEP=det
book	POS=NOUN	DEP=pobj
on	POS=ADP	DEP=prep
Artificial	POS=PROPN	DEP=compound
Intelligence	POS=PROPN	DEP=pobj
.	POS=PUNCT	DEP=punct
Arun	POS=PROPN	DEP=nsubj
helped	POS=VERB	DEP=ROOT
him	POS=PRON	DEP=nsubj
find	POS=VERB	DEP=ccomp
it	POS=PRON	DEP=dobj
.	POS=PUNCT	DEP=punct
Later	POS=ADV	DEP=advmod
,	POS=PUNCT	DEP=punct
they	POS=PRON	DEP=nsubj
discussed	POS=VERB	DEP=ROOT
the	POS=DET	DEP=det
book	POS=NOUN	DEP=dobj
before	POS=ADP	DEP=prep
leaving	POS=VERB	DEP=pcomp
.	POS=PUNCT	DEP=punct

NAMED ENTITIES:

Artificial Intelligence -> ORG

PERSON CANDIDATES:

[]

REFERENCE RESOLUTION:

He -> Arun

him -> Ravi

it -> the book

they -> Ravi + Arun

Resolved paragraph:

Ravi met Arun at the library. Arun was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving.

```
PS C:\Users\lenovo\OneDrive\Documents\DSA0308>
```

Question 2 – Dialog Act Recognition and Response Generation

Conversation: Student: "I submitted my assignment, but I am worried that I may have made many mistakes." Required dialog acts: Reassure + Advise.

Constraints

1. Maintain entity coherence using assignment, mistakes, and you.
2. Use a Cause–Effect or Elaboration relation.
3. Use at least two of: review, improve, confident, feedback.
4. Response must contain 2–3 sentences.
5. Tone must remain positive and polite.
6. Advice must not contradict the reassurance.

Python implementation

```
import spacy

# Install once:
# pip install spacy
# python -m spacy download en_core_web_sm

nlp = spacy.load("en_core_web_sm")

student_input = (
    "I submitted my assignment, but I am worried "
    "that I may have made many mistakes."
)

doc = nlp(student_input)

# Basic linguistic analysis
print("TOKENS:")
print([token.text for token in doc])

print("\nPOS TAGS:")
for token in doc:
```

```
print(f"{token.text:15s} -> {token.pos_}")

# Required dialogue acts
dialog_acts = ["Reassure", "Advise"]

# Entity extraction
entities = [
    chunk.text
    for chunk in doc.noun_chunks
]

print("\nENTITIES / NOUN PHRASES:")
print(entities)

responses = [
    "It is normal to worry after submitting your assignment. "
    "You can review the feedback to improve your work and become more confident.",

    "Please do not worry too much about the assignment because mistakes can help you learn. "
    "Review the feedback so you can improve your work and feel more confident next time.",

    "You have completed the assignment, so take some time to relax. "
    "When feedback arrives, review the mistakes and use them to improve your future work."
]

required_keywords = {
    "review",
    "improve",
    "confident",
    "feedback"
}

def evaluate_response(text):
    response_doc = nlp(text)
```

```

sentence_count = len(list(response_doc.sents))

keyword_count = sum(
    1 for word in required_keywords
    if word in text.lower()
)

polite = any(
    phrase in text.lower()
    for phrase in ["please", "you can", "you have"]
)

positive = any(
    word in text.lower()
    for word in ["normal", "help", "improve", "confident"]
)

return {
    "sentences": sentence_count,
    "required_keywords": keyword_count,
    "length_valid": 2 <= sentence_count <= 3,
    "positive": positive,
    "polite": polite
}

print("\nDIALOG ACTS:")
print(dialog_acts)

print("\nRESPONSE EVALUATION:")
for i, response in enumerate(responses, 1):
    print(f"\nResponse {i}:")
    print(response)
    print(evaluate_response(response))

```

```
print("\nBEST RESPONSE:")
```

```
print(responses[1])
```

```
PS C:\Users\lenovo\OneDrive\Documents\DSA0308> c:: cd 'c:\Users\lenovo\OneDrive\Documents\DSA0308'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '60948' '--' 'c:\Users\lenovo\OneDrive\Documents\DSA0308\WLP_C05AT1 -2.py'
TOKENS:
I
submitted
my
assignment
,
but
I
am
worried
that
I
may
have
made
many
mistakes
.
```

```
POS TAGS:
I                -> PRON
submitted        -> VERB
my               -> PRON
assignment       -> NOUN
,                -> PUNCT
but              -> CONJ
I                -> PRON
am               -> AUX
worried          -> ADJ
that             -> CONJ
I                -> PRON
may              -> AUX
have             -> AUX
made             -> VERB
many             -> ADJ
mistakes         -> NOUN
.                -> PUNCT
```

```
DIALOG ACTS:
['Reassure', 'Advise']

ENTITIES / NOUN PHRASES:
['I', 'my assignment', 'I', 'I', 'many mistakes']

RESPONSE EVALUATION:

Response 1:
It is normal to worry after submitting your assignment. You can review the feedback to improve your work and become more confident.
{'sentences': 2, 'required_keywords': 4, 'length_valid': True, 'positive': True, 'polite': True}

Response 2:
Please do not worry too much about the assignment because mistakes can help you learn. Review the feedback so you can improve your work and feel more confident next time.
{'sentences': 2, 'required_keywords': 4, 'length_valid': True, 'positive': True, 'polite': True}

Response 3:
You have completed the assignment, so take some time to relax. When feedback arrives, review the mistakes and use them to improve your future work.
{'sentences': 2, 'required_keywords': 3, 'length_valid': True, 'positive': True, 'polite': True}

BEST RESPONSE:
Please do not worry too much about the assignment because mistakes can help you learn. Review the feedback so you can improve your work and feel more confident next time.
PS C:\Users\lenovo\OneDrive\Documents\DSA0308>
```

Question 3 – Word Sense Disambiguation and Meaning Representation

Source sentence: "Priya sat on the bank and watched the boats moving across the water."

Possible meanings of 'bank'

Sense	Context fit	Reason
Financial institution	Invalid	The sentence mentions sitting, boats, and water; these do not describe a financial institution.
Side of a river	Valid	A person can sit on a river bank while watching boats move across water.

Predicate logic representation

Let p = Priya, b = the river bank, z = the boats, and w = the water.

```
Person(Priya)
Bank(b)
RiverBank(b)
Water(w)
Boat(z)
sit(Priya, b)
beside(Priya, b)
watch(Priya, z)
move(z, across(w))
on(Priya, b)
```

A compact logical reading is: Priya is a person; b is a river bank; z denotes the boats; w denotes water; Priya sits on/beside b, watches z, and z moves across w.

Unambiguous paraphrase

Priya sat beside the river and watched boats moving across the water.

Python implementation

```
import spacy
import nltk
from nltk.corpus import wordnet as wn

# Install once:
```



```
# pip install spacy nltk
# python -m spacy download en_core_web_sm
# nltk.download("wordnet")
# nltk.download("omw-1.4")

nlp = spacy.load("en_core_web_sm")

sentence = (
    "Priya sat on the bank and watched the boats "
    "moving across the water."
)

doc = nlp(sentence)

print("TOKENS AND POS:")
for token in doc:
    print(f"{token.text:12s} -> {token.pos_}")

print("\nWORDNET SENSES FOR 'BANK':")

bank_synsets = wn.synsets("bank", pos=wn.NOUN)

for synset in bank_synsets:
    print(
        synset.name(),
        "->",
        synset.definition()
    )

# Contextual constraints
context_words = {
    token.lemma_.lower()
    for token in doc
    if not token.is_stop and token.is_alpha
```

```
}

river_context = {
    "boat",
    "water",
    "river",
    "shore",
    "sit"
}

financial_context = {
    "money",
    "loan",
    "account",
    "cash",
    "finance"
}

river_score = len(
    context_words.intersection(river_context)
)

financial_score = len(
    context_words.intersection(financial_context)
)

print("\nCONTEXT SCORES:")
print("River-bank sense:", river_score)
print("Financial-bank sense:", financial_score)

if river_score > financial_score:
    selected = "river-side sense of bank"
else:
    selected = "financial institution sense"
```

```
print("\nSELECTED SENSE:")
```

```
print(selected)
```

```
print("\nPREDICATE LOGIC:")
```

```
print("Person(Priya)")
```

```
print("RiverBank(b)")
```

```
print("Water(w)")
```

```
print("Boat(z)")
```

```
print("sit(Priya, b)")
```

```
print("watch(Priya, z)")
```

```
print("move(z, across(w))")
```

```
print("\nUNAMBIGUOUS PARAPHRASE:")
```

```
print(
```

```
    "Priya sat beside the river and watched "
```

```
    "boats moving across the water."
```

```
)
```

TOKENS AND POS:

Priya	-> PROP
sat	-> VERB
on	-> ADP
the	-> DET
bank	-> NOUN
and	-> CC
watched	-> VERB
the	-> DET
boats	-> NOUN
moving	-> VERB
across	-> ADP
the	-> DET
water	-> NOUN
.	-> PUNCT

WORDNET SENSES FOR 'BANK':

bank.n.01 -> sloping land (especially the slope beside a body of water)
depository financial_institution.n.01 -> a financial institution that accepts deposits and channels the money into lending activities
bank.n.03 -> a long ridge or pile
bank.n.04 -> an arrangement of similar objects in a row or in tiers
bank.n.05 -> a supply or stock held in reserve for future use (especially in emergencies)
bank.n.06 -> the funds held by a gambling house or the dealer in some gambling games
bank.n.07 -> a slope in the turn of a road or track; the outside is higher than the inside in order to reduce the effects of centrifugal force
savings_bank.n.02 -> a container (usually with a slot in the top) for keeping money at home
bank.n.09 -> a building in which the business of banking transacted
bank.n.10 -> a flight maneuver; aircraft tips laterally about its longitudinal axis (especially in turning)

CONTEXT SCORES:

River-bank sense: 3

Financial-bank sense: 0

SELECTED SENSE:

river-side sense of bank

PREDICATE LOGIC:

Person(Priya)

RiverBank(b)

Water(w)

Boat(z)

sit(Priya, b)

watch(Priya, z)

move(z, across(w))

UNAMBIGUOUS PARAPHRASE:

Priya sat beside the river and watched boats moving across the water.

PS C:\Users\lenovo\OneDrive\Documents\DSA0308>